

BDMM 2026: HW Answers

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Database Schema Transformation and Design Decisions

The database underwent a series of transformations aimed at improving data consistency, reducing redundancy, and enhancing query performance through schema redesign and indexing.

Initially, exploratory analysis revealed that the dataset followed a heavily embedded document model, particularly for host and review data. Host information was stored redundantly within each listing document, despite the same host appearing multiple times across listings. This was confirmed through host ID frequency analysis, which showed repeated occurrences of identical `host_id` values. Additionally, review data was embedded as arrays within listings, leading to potentially large and unbounded document sizes.

To address these issues, the schema was refactored from an embedded to a referenced data model. Host data was extracted into a separate `hosts` collection, with each host stored only once and identified by a unique `host_id`. Listings were modified to retain only a reference (`host_id`) instead of the full embedded host document. Similarly, review data was extracted into a dedicated `reviews` collection, where each review is stored as an individual document and linked back to its corresponding listing via `listing_id`. This restructuring follows a referencing pattern and enables more efficient storage, improved maintainability, and greater flexibility in querying.

Further data cleaning steps were performed to remove redundant and non-informative fields. The top-level `review_scores_rating` field was removed, as it contained only null values, while the nested `review_scores.review_scores_rating` field was retained as the valid source of rating information. The `transactions` field was also removed due to its uniform and non-informative structure.

To support the new referenced schema, indexes were introduced on key fields used for relationships between collections. A unique index was created on `hosts.host_id` to enforce entity integrity, while non-unique indexes were added on `listings.host_id` and `reviews.listing_id` to optimize join operations and improve query performance. Additional compound indexes were later introduced during query optimization to further enhance execution efficiency.

Overall, the transformation replaced a redundant embedded structure with a normalized, reference-based design, reduced unnecessary data duplication, simplified document structure, and significantly improved the efficiency of data retrieval operations.

Part A

A1. Price Analysis

The analysis is based on a standardized scenario in which a stay duration of seven nights and three guests is assumed. It focuses on listings located in Hong Kong, Montreal, and Barcelona, and considers five property types: Apartment, Guesthouse, Hostel, House, and Townhouse. To ensure that the selected listings are suitable for this scenario, only properties with `accommodates` ≥ 3 are included, as they must be able to host three people.

Availability is also taken into account. Listings must have `availability_365` ≥ 7 to ensure that a seven-night stay is possible within the next year. In addition, booking rules must allow such a stay, meaning that `minimum_nights` $\leq 7 \leq$ `maximum_nights`. All these conditions were used to approximate that the selected properties are realistically bookable for the defined scenario. However, this requirement cannot be fully guaranteed without access to calendar data, which would be needed to verify that seven nights are available consecutively.

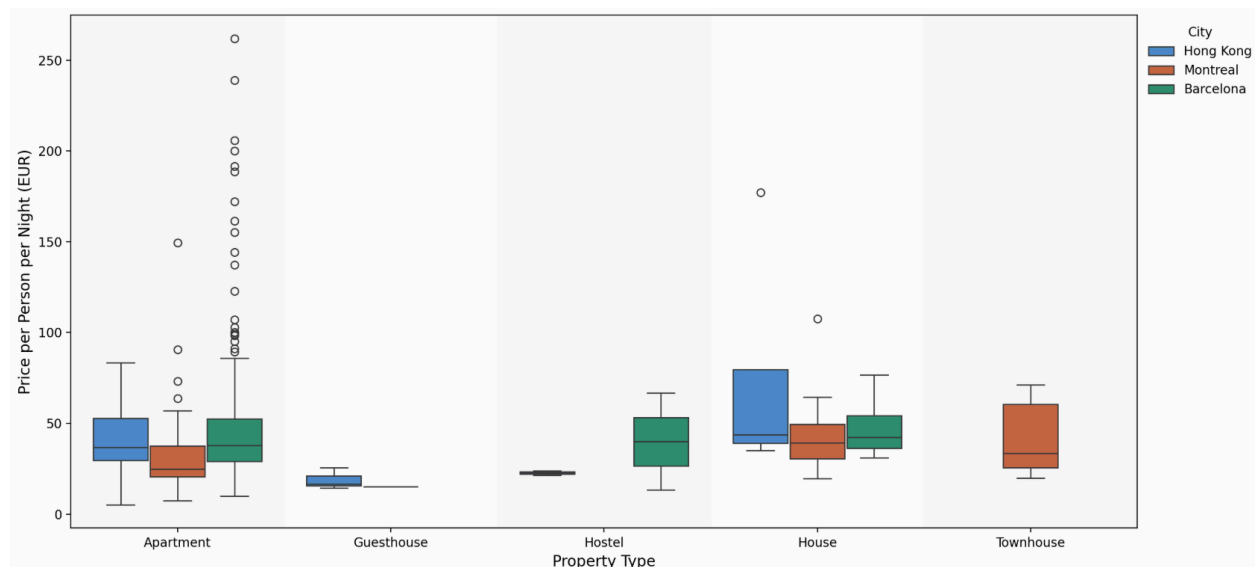
Price calculations rely only on structured pricing fields. When available, weekly_price is compared to the cost of seven nightly prices ($\text{price} \times 7$), and the lower value is used as the base accommodation cost. If the nightly price is missing, the weekly price is used directly. This approach ensures that any weekly discounts are consistently incorporated. Additional charges mentioned in unstructured text fields are excluded, as they are inconsistently recorded and may not always represent actual costs.

The total price consists of three components: the base accommodation cost, the cleaning_fee, and any extra person charges. Extra charges are calculated using the fields guests_included and extra_people. Specifically, the number of extra guests is defined as $\max(0, 3 - \text{guests_included})$, and the corresponding cost is $\text{extra_guests} \times \text{extra_people} \times 7$.

Missing values are handled conservatively: cleaning_fee and extra_people are set to zero if absent, while guests_included is assumed to be one, reflecting that a listing typically includes at least one guest in the base price.

Finally, since prices are recorded in different currencies (HKD, CAD, EUR), all values are converted to euros to ensure comparability across cities.

**Figure 1. Price per Person per Night
(3 people, 7 nights)**



The chart presents price distributions per person per night across five property types, with each group containing three boxplots corresponding to Hong Kong, Barcelona, and Montreal. Although the assignment refers to “rating categories,” these are not defined,

unlike property types, which are explicitly specified. Moreover, the chart already incorporates two categorical dimensions (property type and city), resulting in 15 boxplots. Introducing rating categories would require a third categorical dimension, which cannot be represented within a single two-dimensional boxplot. Therefore, property types are used as the primary grouping variable.

Prices vary across both cities and property types. Boxplots summarize these differences using medians, quartiles, and whiskers, allowing comparison of central values, variability within the middle 50% (IQR), and the presence of extreme values.

Apartments are the most informative category due to the largest number of observations. Montreal has the lowest median apartment price (24.78 euros) and the narrowest IQR (approximately 20.57–37.49), indicating lower and more consistent pricing. Barcelona and Hong Kong show similar median levels (37.67 and 36.90 euros), but wider IQRs, suggesting greater variability. Barcelona also exhibits the widest overall range, with prices extending from about 9.76 to 261.90 euros.

The relationship between whiskers and maximum values highlights high-price listings. In Hong Kong, the upper whisker coincides with the maximum (83.32 euros), indicating no outliers. In contrast, Barcelona includes values beyond the upper whisker, extending above 260 euros, suggesting a premium segment rather than isolated anomalies.

Other property types have limited observations after filtering, reducing the reliability of comparisons. Overall, apartments provide the most robust insights, showing lower and more stable prices in Montreal and greater variability, including high-end listings, in Barcelona.

A2. Query Pipeline Analysis

The analysis is based on a standardized scenario in which a stay duration of seven nights and three guests is assumed. It focuses exclusively on listings located in New York. To ensure that selected properties are suitable for this scenario, only listings that can accommodate at least three guests are included. Availability is also considered: listings must have at least seven available days within the next year ($\text{availability_365} \geq 7$), and their booking rules must allow a seven-night stay, meaning that $\text{minimum_nights} \leq 7 \leq \text{maximum_nights}$.

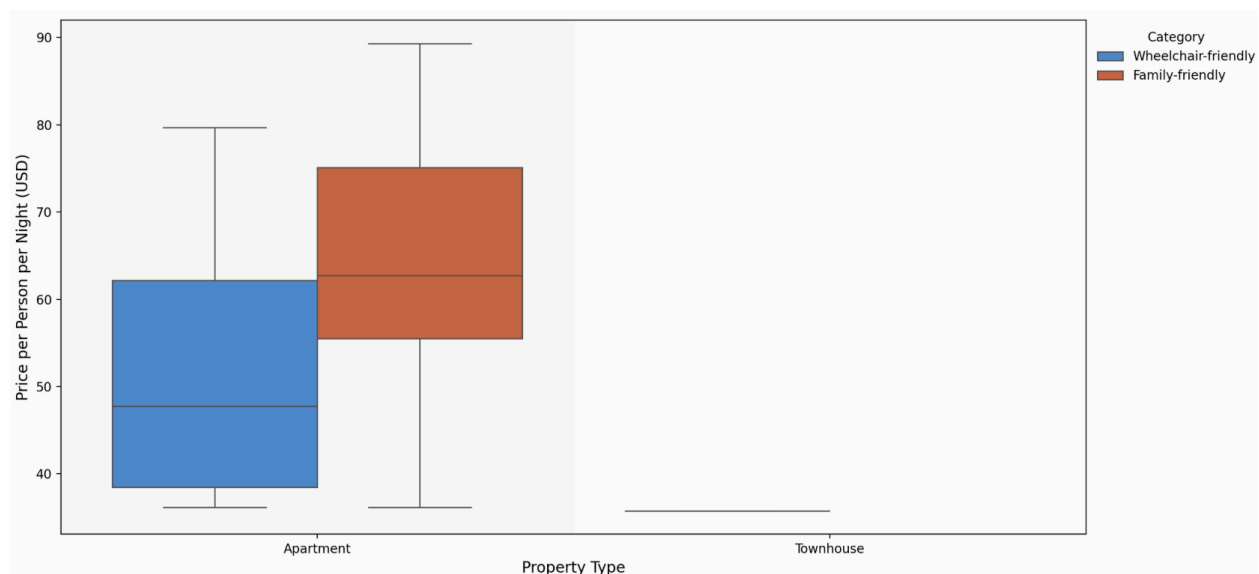
Only listings that contain a specific set of required amenities are included, namely Wifi, Hair dryer, 24-hour check-in, Air conditioning, and Bed linens. Amenities are matched exactly as written in the dataset; therefore, similar but not identical entries such as “Self check-in,” “Lockbox,” or “Keypad” are excluded. Listings are further categorized as

wheelchair-friendly if they include at least one accessibility-related amenity (such as Elevator or Wheelchair accessible), and as family-friendly if they include at least one family-oriented feature (such as Crib or Family/kid friendly). A listing may belong to both categories simultaneously. Due to the strict filtering conditions, fewer than five property types may remain; in such cases, all available types are used rather than enforcing a fixed number.

Price calculations rely only on structured pricing fields. When available, the weekly_price is compared to the cost of seven nightly prices, and the lower value is used to reflect potential discounts. If the nightly price is missing, the weekly price is used directly. The total price includes the base accommodation cost, cleaning_fee, and any additional charges for extra guests. For three guests staying seven nights, extra charges are calculated based on the number of guests not included in the base price and the per-night extra guest fee.

Missing values are handled conservatively: cleaning_fee and extra_people are set to zero if absent, while guests_included is assumed to be one. Listings without any usable price information are excluded from the analysis. Since categories may overlap, the same property types are used for comparison between wheelchair-friendly and family-friendly listings.

Figure 2. Price per Person per Night by Property Type and Category (3 people 7 nights)



Strict filtering criteria—such as locality (New York), availability, required amenities, and accessibility—significantly reduce the number of listings. As a result, only a few property

types remain: apartments for family-friendly listings, and apartments and townhouses for wheelchair-friendly listings.

Apartments allow meaningful comparison, as they appear in both categories. Median prices are higher for family-friendly apartments (62.71) than for wheelchair-friendly ones (47.76). The interquartile range is slightly narrower for family-friendly apartments (55.50–75.07) compared to wheelchair-friendly ones (38.44–62.17), suggesting somewhat lower variability for family-friendly apartments in the central 50% of values. Family-friendly apartments also reach a higher maximum (89.29 versus 79.67), indicating a broader upper price range.

Townhouses appear only once, so no variability can be observed.

Table 1. Query Execution Statistics

	stage_type	execution_time_ms	documents_examined	keys_examined	memory_usage
version					
before_optimization	COLLSCAN	28	5555	0	N/A
1st_optimization_idx_address_market	IXSCAN	18	607	607	N/A
2nd_optimization_compound_idx_market_availability	IXSCAN	7	379	379	N/A
3rd_optimization_compound_idx_market_availability_accomodates	IXSCAN	7	175	315	N/A
4th_optimization_compound_idx_market_availability_accomodates_min_nights	IXSCAN	6	146	314	N/A

The table demonstrates the impact of indexing on query execution efficiency, highlighting changes in execution time, documents examined, and keys examined across successive optimization stages.

Initially, the query performs a full collection scan (COLLSCAN), examining all 5,555 documents with no index usage, resulting in the highest execution time (28 ms). Introducing an index on address.market significantly improves performance by restricting the search space to New York listings, reducing both documents and keys examined to 607 and lowering execution time to 18 ms.

Further improvements are achieved through compound indexing. Adding availability-related fields reduces both documents and keys examined to 379, with execution time decreasing to 7 ms, indicating a more selective and better-aligned index. In later stages, the number of examined documents continues to decrease—from 175 to 146—while the number of keys examined remains relatively high (315 and 314). This shows that although document retrieval becomes more efficient, the query still requires scanning a larger number of index entries to locate matching results. Importantly, the filtering conditions and final result set remain unchanged; only the efficiency of accessing the data is improved.

The final optimization achieves the lowest execution time (6 ms) and the smallest number of examined documents (146), but only marginal improvement in keys examined, indicating diminishing returns from further index refinement. Memory usage is reported as N/A, as it is not provided for this query. Overall, indexing substantially enhances performance by reducing the volume of processed data, while later optimizations primarily refine document access rather than index traversal.

A3. Analytical Price Suggestion

The analysis focuses on listings in Barcelona, Hong Kong, and Montreal, and considers five property types: Apartment, Guesthouse, Hostel, House, and Townhouse. The objective is to predict missing price values using similar listings with known prices. Comparisons are restricted to listings within the same city, as prices are recorded in different currencies and reflect different market conditions.

Only listings with sufficient information are used. Specifically, each listing must have city, property_type, room_type, and accommodates. Listings missing these fields are excluded, and extreme values are filtered out to ensure realistic comparisons.

Prices are standardized using price per person per night in local currency, defined as $\text{ppn_local} = \text{price} / \text{base_guests}$. The final suggested price is calculated as $\text{price_suggestion} = \text{predicted ppn_local} \times \text{base_guests}$. The number of base guests is defined as guests_included when available and greater than zero; otherwise, it is assumed to be one. This reflects the idea that base pricing is typically structured around included guests rather than maximum capacity.

Similarity between listings is determined using both numerical and categorical features. Numerical features include accommodates, bedrooms, bathrooms, beds, review_score_avg, and minimum_nights, while categorical features include property_type and room_type. The variable review_score_avg is used as a proxy for guest sentiment, approximating the use of sentiment analysis on review text without directly processing it. Categorical variables are one-hot encoded, and all features are standardized before applying the K-nearest neighbors method.

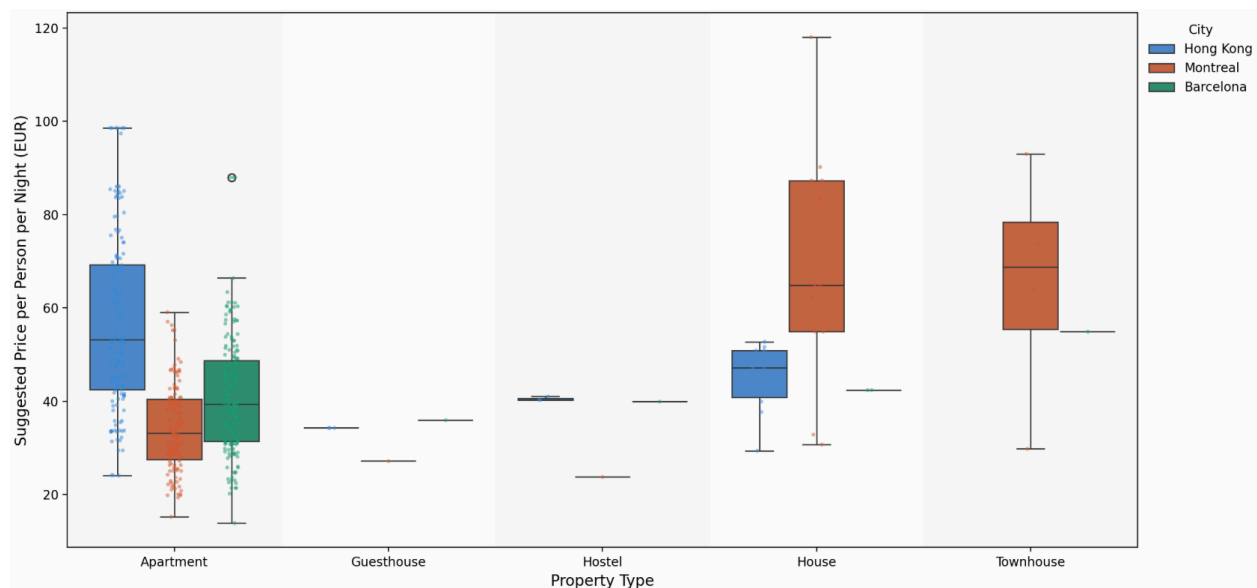
Outliers are removed within each city and property type group based on price per person per night. Smaller groups are handled using simple ratio or median-based rules, while larger groups use the standard IQR method.

Predictions are generated using up to five nearest neighbors. A fallback strategy is applied: first using neighbors with the same city, property type, and room type; then

relaxing to the same city and property type; and finally using the median of all neighbors if necessary.

Missing numerical values are set to zero and missing categorical values to “MISSING”
The final suggested prices are converted to euros for consistent visualization.

Figure 3. Suggested Prices Per Person per Night by Property Type and City



The predicted price distributions broadly follow the structure of the observed prices across Barcelona, Hong Kong, and Montreal (see), with the most reliable insights coming from apartments due to their substantially larger number of observations. Apartments dominate the dataset (171 in Barcelona, 148 in Hong Kong, and 130 in Montreal), which makes their distributions more stable and representative. For Barcelona, the predicted median price is approximately €39.4, with an interquartile range (IQR) of about €17.4 (Q1 ≈ €31.35, Q3 ≈ €48.73), closely matching the observed structure. Similarly, in Montreal, the predicted median (€33.1) and IQR (€12.9) remain reasonably aligned with the original data, although the spread is slightly more compressed, suggesting a modest smoothing effect of the model.

In contrast, Hong Kong exhibits higher predicted values, with a median of approximately €53.2 and a wider IQR of €26.9 (Q1 ≈ €42.45, Q3 ≈ €69.31). This indicates a noticeable upward bias compared to observed prices. The likely explanation lies in data limitations

and the fallback strategy: when sufficiently similar listings are not available, broader matching criteria introduce neighbors with higher prices, which inflates predictions.

For other property types—guesthouses, hostels, houses, and townhouses—the analysis is constrained by small sample sizes. Many categories contain only one or a few observations (e.g., guesthouses and hostels are represented by only 1–3 listings with predicted prices), resulting in zero or near-zero IQRs and limited variability. Houses show more variation, particularly in Montreal (median $\approx$ €64.9, IQR $\approx$ €32.4), but still remain much less robust than apartments. Townhouses are also sparse, with only a few observations available.

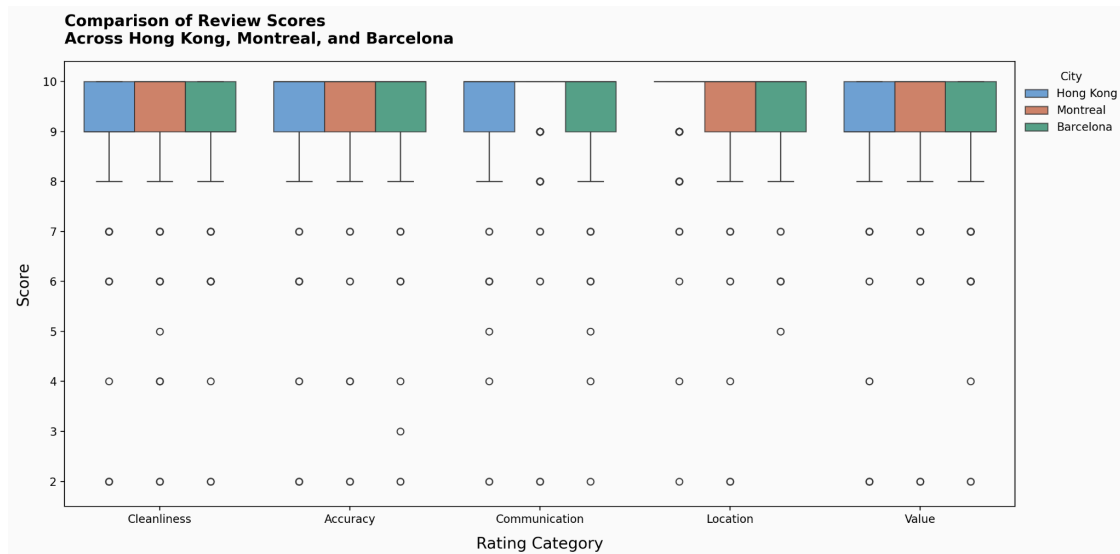
Despite these limitations, predicted values generally fall within plausible ranges across all cities and property types. In practice, all listings with missing prices received a prediction. The strongest and most reliable performance is observed for apartments, while deviations in other categories are primarily explained by data scarcity and the necessity of fallback logic.

Part B

B1 Interpretation of Review Score Distributions

Missing values in the review score dimensions were interpreted as cases where users chose not to provide a category rating, rather than indicators of poor performance or data quality issues. These values were excluded from the analysis to avoid bias, ensuring only observed ratings contributed to the distributional assessment. No additional inconsistencies were identified, as all recorded values fell within the expected range with no signs of corruption or invalid formatting. The analysis of review scores across Hong Kong, Montreal, and Barcelona reveals a highly consistent pattern of strongly positive evaluations across all five dimensions: cleanliness, accuracy, communication, location, and value (figure 4).

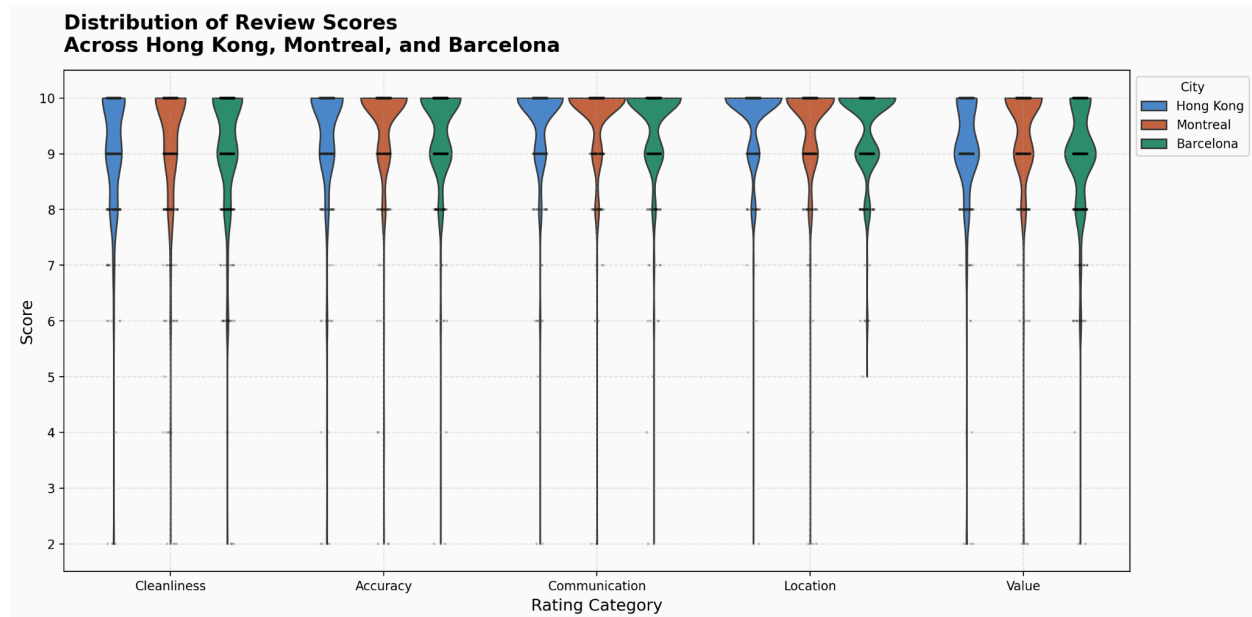
Figure 4. Distribution of Review Scores Across Rating Categories in Hong Kong, Montreal, and Barcelona



Upon observing the box plots, the distributions are strongly compressed toward the upper bound, with upper whiskers largely absent due to the convergence of the third quartile (Q3) and the maximum value. Variability is almost entirely concentrated in the lower tail, evidenced by outliers ranging approximately from 2 to 7, suggesting that while most listings perform exceptionally well, a small subset receives substantially lower evaluations. Differences across cities remain subtle. Communication and location appear marginally more stable, exhibiting slightly higher consistency and fewer extreme deviations, while cleanliness and value display somewhat greater dispersion in lower-end outliers. Overall, all three cities demonstrate broadly similar distributions, indicating a consistently high standard of service across markets.

A key limitation of the rating scale lies in its restricted discriminative power. The pronounced clustering near the upper bound implies that minor differences in ratings carry disproportionate informational weight, thereby limiting the metric's ability to meaningfully distinguish among high-performing listings. To address this limitation, a violin plot combined with a strip plot was employed. This visualization more effectively reveals the underlying distribution densities while also highlighting individual observations across discrete rating values.

Supplementary Figure 4. Problem-Specific Visualization Addressing Box Plot Limitations in the Distribution of Review Scores

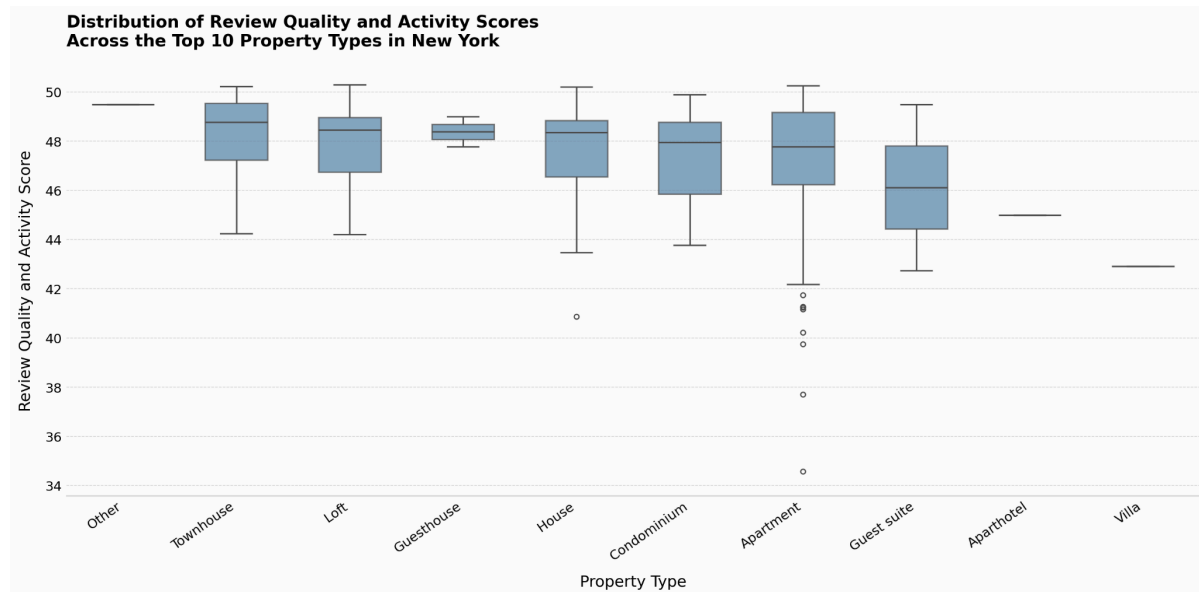


B2 Score Analysis Across Top 10 Property Types & Database Performance Analysis

Score across top 10 property types in New York

Review Quality and Activity Scores are relatively high and tightly concentrated across most property types, with median values ranging between approximately 46 and 49 and few outliers. The exception is Apartments, where a subset of listings significantly underperforms; an expected finding given that Apartments represent the dominant property type, making outliers statistically more likely. Property types such as Villa, Aparthotel, and Other show no interquartile range, which is explained by their sample size of a single listing each, rendering any distributional analysis for these categories statistically meaningless.

Figure 5. Distribution of Review Quality and Activity Scores Across Top Property Types in New York



Indexing impact on query execution

The pipeline's optimization prioritized indexing on the most computationally expensive fields first (address.market, listing_id), followed by less intensive but relevant fields (property_type, date).

Table 1. Query Execution Statistics

version	stage_type	execution_time_ms	documents_examined	keys_examined	memory_usage
taskB2_before	COLLSCAN	26,063	5,555	0	51,5912
taskB2_idx_listing	COLLSCAN	64	5,555	0	51,5912
taskB2_idx_market	IXSCAN	25,791	607	607	51,5912
taskB2_idx_market_property_type	IXSCAN	26,040	607	608	51,5912

taskB2_idx_market_property_type_l ast_review	IXSCAN	26,066	538	544	51,5912
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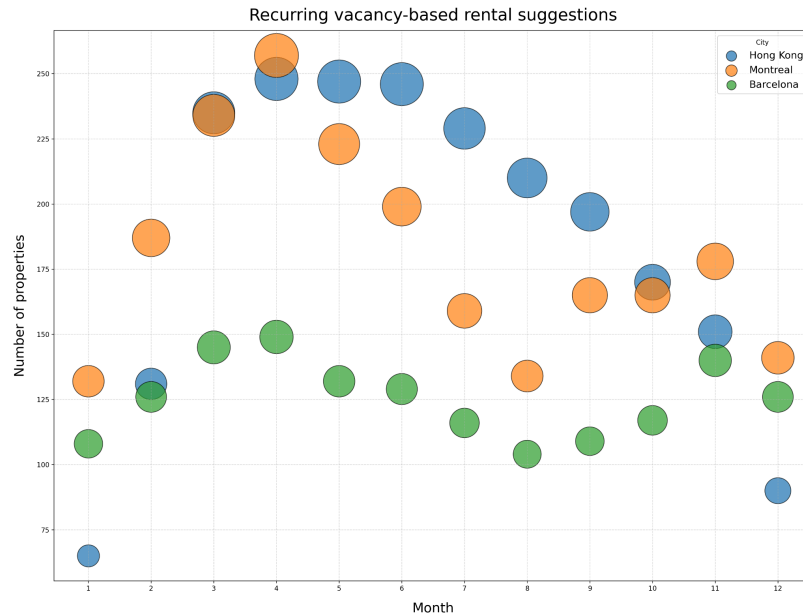
A baseline execution without indexes produced a full collection scan (COLLSCAN) over all 5,555 documents. Indexes were introduced progressively based on query structure and field cardinality, ordering from lower to higher cardinality to maximize filtering efficiency: address.market first, followed by `property_type`, then `last_review`. The address.market index reduced examined documents from 5,555 to 607 and enabled an IXSCAN. Adding `property_type` yielded no further selectivity gain, while including last_review reduced examined documents to 538. An index on `reviews.listing_id` was tested to optimize the \$lookup but had no visible effect on the top-level execution plan, which remained a COLLSCAN. Execution time was also inconsistent due to caching effects. The compound index (`address.market`, `property_type`, `last_review`) proved the most effective optimization strategy.

B3 Interpretation of recurring vacancies

The year of each review was extracted and the time difference between consecutive reviews computed. A total of 11,183 missing values were observed in the `gap_days` variable, corresponding to the first review entry within each (`listing_id`, `year`) group, where no prior observation exists. These were considered structurally expected and replaced with zero. To identify vacancy periods, the data was reorganised into a complete listing-year-month grid, ensuring all months were represented regardless of review activity. Only sequences of more than one consecutive month without reviews were retained, as these reflect meaningful inactivity. Patterns were further filtered to include only those recurring consistently across multiple years for the same listing, ensuring recommendations reflect sustained rather than isolated vacancies. The corresponding months were stored in the `rental_suggestion` field.

Vacancy patterns were aggregated at the city level and are presented in Figure 6, which displays the number of properties suitable for long-term rental per month, with marker size reflecting property volume and colour distinguishing each city.

Figure 6. Monthly Distribution of Vacancy-Based Rental Availability Across Hong Kong, Montreal, and Barcelona



Across all three cities, vacancy is not uniformly distributed, indicating seasonal demand variation. In Hong Kong, suitable properties peak between March and June, with the highest concentration in April, suggesting reduced short-term occupancy during spring. A secondary decline appears in December, pointing to a rebound in short-term demand over the holiday period. Montreal exhibits a similar but slightly shifted pattern, peaking in April and remaining elevated through May and June before declining over summer; consistent with increased tourism and short-term demand reducing long-term availability. A moderate secondary opportunity window emerges in early autumn. Barcelona shows a more stable but lower overall vacancy level. A mild peak occurs around March–April, though monthly variation is less pronounced, reflecting more consistent year-round short-term demand. Overall, spring months (March–May) represent the most consistent opportunity for long-term rental strategies across all cities. The recurring nature of these gaps strengthens the reliability of these findings. Hosts could optimise occupancy and revenue by targeting long-term tenants during predictable low-demand periods, particularly in markets with stronger seasonal variation such as Hong Kong and Montreal.

Part C

C1 Amenity Premium Analysis

Assumptions

For C1, I assumed a stay of 7 nights for 2 guests, as required. To make prices comparable across listings, I calculated an effective total stay cost that includes the lower of 7 x nightly price and weekly_price when a weekly rate exists. Then, I added cleaning_fee and any extra_people charge when the target of 2 guests exceeds guests_included. Missing cleaning_fee and extra_people were treated as 0, while missing guests_included was treated as 1.

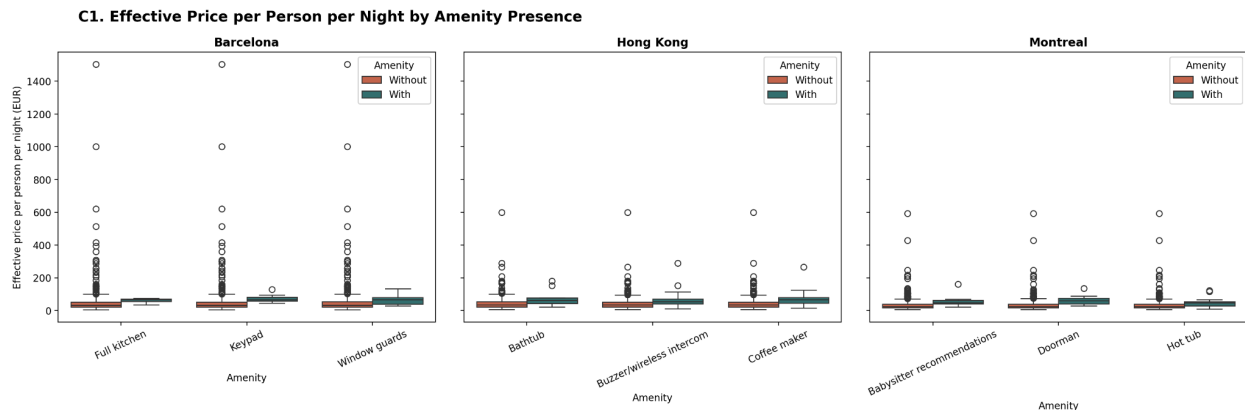
Since the three cities use different currencies, the final effective price per person per night was converted into EUR. This was done to improve cross-city visual comparability; the ranking of amenities inside each city is still based on within-city comparisons.

The amenity effect was measured within each city, not across cities, because Hong Kong, Montreal and Barcelona have different market structures. For each amenity, listings were split into two groups: with the amenity and without the amenity. Then, the median effective price per person per night of the two groups were compared. Median was preferred over mean because Airbnb prices are often skewed by extreme luxury listings and the median gives a more robust central estimate.

Amenities present in every listing were excluded because they provide no variation. A stability rule was implemented to require at least 5 listings with an amenity before considering it in the ranking, so that rare amenities would not dominate the results.

The results should be interpreted as association, not causation. If an amenity is linked to higher prices, this does not prove that the amenity alone causes the increase. It may instead signal a broader premium property profile, better location, or higher overall host quality.

Visual



As displayed above, the x-axis is organized by amenity within each city, and the y-axis shows the effective price per person per night in EUR. For each amenity, the two side-by-side box plots compare listings that include the amenity against those that do not.

It shows not only the median price difference, but also the spread of prices, including the interquartile range, minimum, and maximum. In Hong Kong, the strongest price-premium amenities are Coffee maker, Bathtub and Buzzer/wireless intercom. In Montreal, they appear to be Doorman, Hot tub and Babysitter recommendations. In Barcelona, the top amenities are Window guards, Full kitchen and Keypad.

The main pattern in the chart is that the “with amenity” box usually sits above the “without amenity” box, meaning listings offering these amenities tend to have higher median effective prices. In several cases, the boxes for the “with” group are also wider, which suggests that higher-end listings are more heterogeneous in price. Overall, the figure visually supports the idea that certain amenities are associated with more expensive listings, although the strength of the effect varies by city.

Interpretations

For C1, the results suggest that amenity effects on price are real, but they are not uniform across cities. In all three markets, the strongest positive “price premium” amenities are those that likely signal a higher overall property standard rather than just a single extra feature. In Hong Kong, the clearest premium is associated with coffee maker, bathtub and buzzer/wireless intercom, while in Montreal the largest median differences appear for doorman, hot tub and babysitter recommendations. In Barcelona, window guards, full kitchen and keypad show the strongest upward shift in effective price per person per night.

The box plots show that listings with these amenities generally have higher medians, and in many cases also a wider interquartile range. This indicates not only higher central prices, but also greater heterogeneity within the more highly equipped segment. In other words, these amenities tend to appear more often in listings that are already positioned as mid-range or premium offers. The results should therefore be interpreted as an association with higher prices, not proof that adding one amenity alone will cause the full observed increase.

At the same time, some amenities appear to correlate with lower prices. This likely reflects market positioning rather than negative value. Amenities that are common in simpler, budget-oriented, or more compact listings may be associated with lower medians because they are typical of lower-priced accommodation segments. Therefore, an amenity can be useful to guests without being linked to a price premium.

Across cities, the ranking of top amenities is not perfectly stable, which suggests local market differences in what guests are willing to pay more for. However, the broader pattern is consistent: amenities that improve convenience, comfort, or perceived quality tend to be associated with higher prices, while more basic or widely expected features contribute less to price differentiation. Overall, C1 shows that amenities can serve as a meaningful pricing signal, especially when they reflect a broader upscale listing profile.

C2

Assumptions

For C2, the analysis was restricted to Sydney and New York and only to the specified comparable property types: Aparthotel, Apartment, Condominium, Loft, and Serviced apartment.

Only listings with at least 30 reviews were included since review-based quality signals are less reliable for listings with very small sample sizes. “Top-rated” was defined separately within each city using the 85th percentile of `review_scores_rating`, rather than one global cutoff, because rating distributions can differ by city. In this dataset, the cutoff was by coincidence 98.0 in both Sydney and New York.

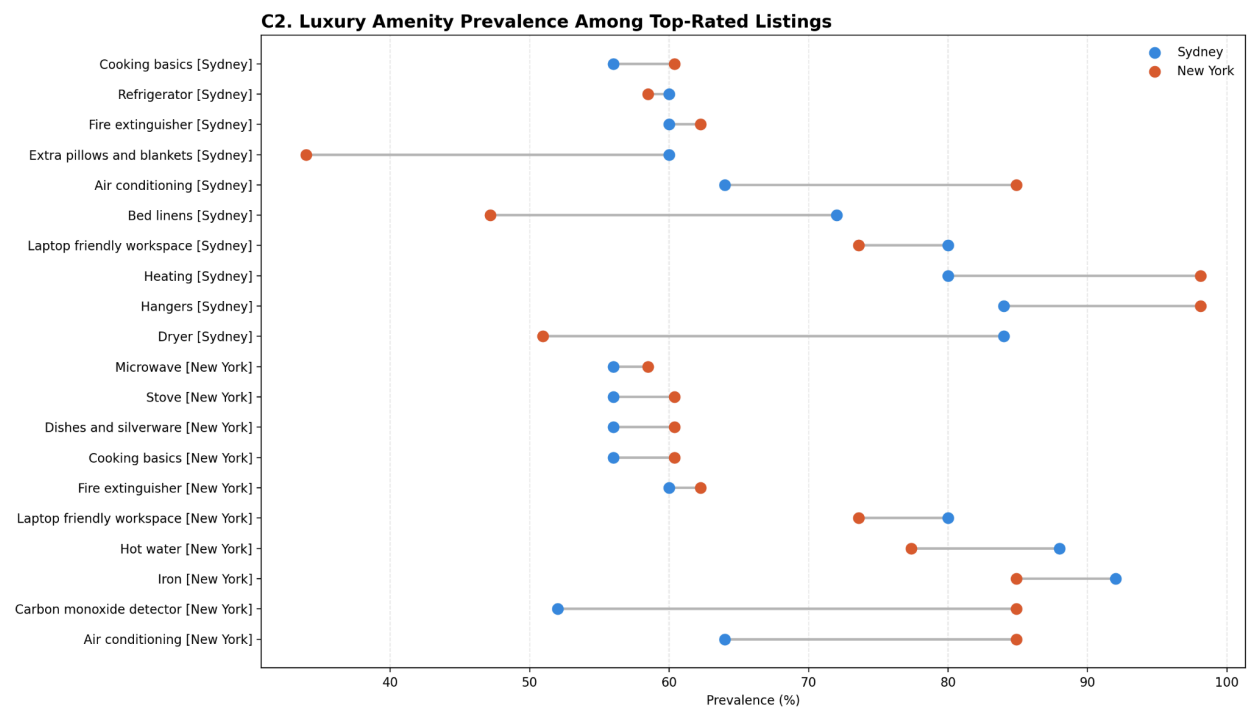
For each top-rated subset, amenity prevalence was calculated as:

$$\text{Prevalence} = 100 \times (\text{Listings with amenity} / \text{Total top-rated listings in that city})$$

Amenities appearing in at least 85% of top-rated listings were treated as essential rather than luxury as near-universal features no longer distinguish premium listings. All remaining amenities were classified as optional luxury amenities, and only these were ranked.

For the performance analysis, I evaluated only the aggregation pipeline used to select the relevant listings. The “before” version used the same logic without the added compound index, while the “after” version added the index on address.market, property_type, number_of_reviews, and review_scores.review_scores_rating. Performance was compared using explain("executionStats").

Visual



The double lollipop chart compares the prevalence of the top 10 optional luxury amenities among top-rated listings in Sydney and New York. Each amenity is shown with two color-coded markers, blue for Sydney and red for New York, allowing direct visual comparison of how common that amenity is in each city.

The results show that Sydney’s leading luxury amenities are Dryer and Hangers at 84.0%, followed by Heating and Laptop friendly workspace at 80.0%. In New York, the most prevalent luxury amenities are Air conditioning, Carbon monoxide detector and

Iron, each at 84.91%, followed by Hot water at 77.36% and Laptop friendly workspace at 73.58%.

This suggests that both cities share a strong comfort baseline, but the specific optional amenities that distinguish top-rated listings differ. Sydney appears to emphasize practical residential comfort features, while New York gives greater importance to indoor climate and safety-related amenities. The chart therefore shows that “luxury” is partly city-specific rather than universal.

Visual of query before and after

version	stage_type	execution_time_ms	documents_examined	keys_examined	memory_usage
before_optimization	IXSCAN	35	1216	1218	None
after_optimization	IXSCAN	20	346	357	None

As displayed in the table above, the added compound index made the scan much more selective in the “after” version. Before optimization, the pipeline took 35 ms, examined 1216 documents and examined 1218 keys. After optimization, execution time fell to 20 ms, documents examined dropped to 346 and keys examined fell to 357.

Interpretations

The C2 analysis shows that top-rated properties in Sydney and New York share many essential amenities, while differing more clearly in their optional luxury features. This means highly rated listings in both cities meet a common baseline of comfort, while the amenities that distinguish them most are more city-specific.

In Sydney, the most prevalent luxury amenities are dryers, hangers and heating, whereas in New York the leading optional amenities are air conditioning, carbon monoxide detector and iron. The double lollipop chart highlights these contrasts well: amenities with similar prevalence in both cities suggest a shared premium standard, while large gaps point to differences in local housing style, guest expectations, or market positioning.

The results also suggest that “luxury” is not universal. Some amenities appear strongly tied to one city, which may reflect local building characteristics or traveler preferences rather than overall quality alone.

From a query-performance perspective, the optimized version should reduce documents examined and improve execution time by filtering more efficiently on city, property type, review count, and rating. Overall, the findings show that top-rated listings combine a stable essential foundation with a smaller set of optional amenities that vary meaningfully between Sydney and New York.

C3

Assumptions

For C3, once again I assumed a 7-night stay for 2 guests and used an effective total cost that includes the cheaper of nightly total versus weekly_price, plus cleaning_fee and any applicable extra_people charge. Missing fee fields were treated as 0, and missing guests_included as 1.

To measure guest-perceived value, a custom weighted metric was created:

$$\text{weighted_value_score} = 0.60 \times \text{value_rating_score} + 0.30 \times \text{affordability_score} + 0.10 \times \text{review_confidence_score}$$

Here, $\text{value_rating_score} = \text{review_scores_value} / 10$, $\text{affordability_score}$ is the inverse min-max normalized price per person per night within each city, and $\text{review_confidence_score} = \min(\text{number_of_reviews} / 50, 1)$.

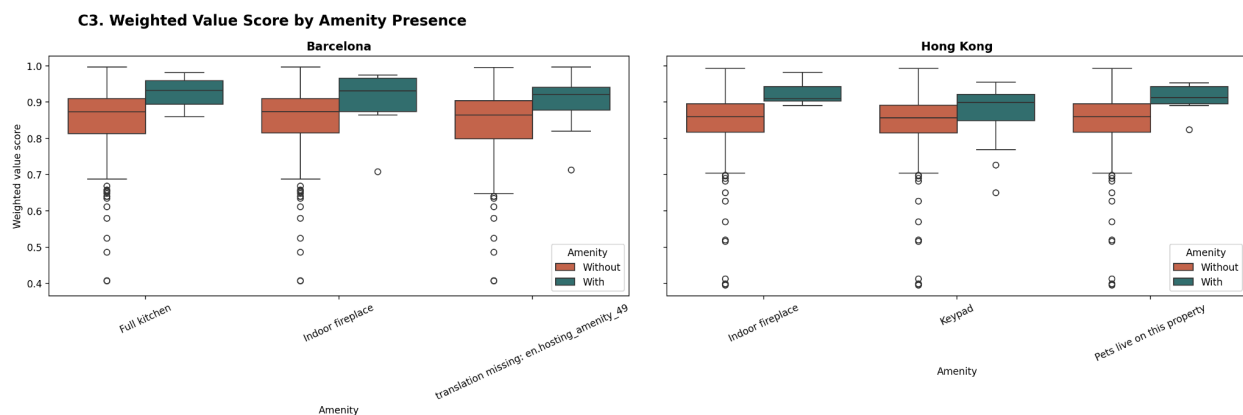
The highest weight was given to $\text{review_scores_value}$ since it directly reflects how guests judge value for money. Affordability was given a substantial but smaller weight so that lower cost still matters. Review count received a small weight to reduce the influence of listings with very few reviews.

Normalization was done within each city rather than across cities, because Hong Kong and Barcelona have different price levels and direct raw-price comparison would be misleading. Amenities present in more than 85% of listings were again excluded for the

same reasons mentioned previously and also at least 5 listings with an amenity were required before measuring its effect.

Amenity impact was measured by the difference in median weighted value score between listings with and without each amenity. As in C1, this should be interpreted as association rather than proof of causality. Some amenities may signal better hosting quality or more complete accommodation rather than independently causing higher perceived value.

Visual



Unlike C1, here the y-axis is not price, instead it shows the custom `weighted_value_score`, which combines guest value ratings, affordability, and review confidence. The figure compares the top 3 amenities associated with higher weighted value in Barcelona and Hong Kong, again using paired box plots for listings with and without each amenity.

For Barcelona, the strongest positive amenities are Full kitchen, Indoor fireplace and *translation missing: en.hosting_amenity_49* [1]. For Hong Kong, they are Pets live on this property, Indoor fireplace and Keypad. In each case, the visual compares whether listings that include the amenity tend to achieve higher weighted value scores than those that do not.

The main visual pattern is that the “with amenity” boxes are generally shifted upward relative to the “without amenity” boxes. This means those amenities are associated with higher perceived value, not simply higher prices. Because the weighted score rewards both satisfaction and affordability, an upward shift suggests that guests may see these features as improving value for money. The chart therefore helps distinguish amenities linked to stronger guest-perceived value from those that may only signal higher cost. As a result, C3 supports the later recommendation step, where below-median listings are advised to add amenities that are associated with better value scores.

Interpretations

The C3 analysis suggests that guest-perceived value is shaped by more than low price alone. The weighted metric gives the greatest importance to `review_scores_value`, while still rewarding affordability and giving a smaller adjustment for review volume. This makes the final score more balanced than using price or ratings in isolation. A listing can therefore perform well by combining strong value ratings with reasonable effective cost per person per night, rather than simply being the cheapest option.

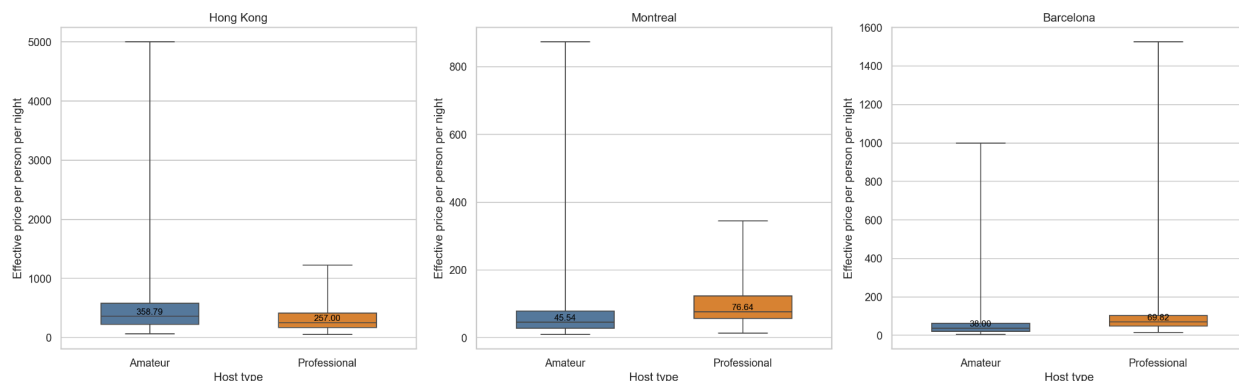
For Barcelona and Hong Kong, the most impactful amenities are those associated with higher median weighted value scores when present than when absent. This implies that some amenities improve how guests judge the relationship between quality and price, not just the absolute standard of the property. At the same time, the strongest amenities are not necessarily identical across the two cities, which suggests that local market expectations matter. What guests perceive as “good value” in one city may not carry the same effect in another.

The recommendation table focuses on below-median listings and proposes amenities that are currently missing but are positively associated with higher weighted value. These suggestions are practical because they target features linked to better perceived value rather than only higher prices. However, the results remain associative rather than causal: an amenity may be acting as a signal of a broader hosting standard, better maintenance, or a more complete guest experience. Overall, C3 indicates that improving perceived value likely depends on a combination of fair pricing and a thoughtful amenity package tailored to each city.

Part D

D1 Host Type and Pricing

D1. Effective Price Per Person Per Night by Host Type and City



(Graph: Box plots of `price_per_person_per_night` by `host_type` across cities)

This analysis examines how host type influences pricing in Hong Kong, Montreal, and Barcelona. Hosts were classified using `host_id` and listing counts into **Professional (multiple listings)** and **Amateur (single listing)**. Pricing was measured using `price_per_person_per_night`, calculated from `price`, `weekly_price`, `cleaning_fee`, `security_deposit`, and `extra_people`, assuming two guests over seven nights.

The box plot shows that pricing behaviour varies significantly by city. In Barcelona, the median price for professional hosts is **69.82**, compared to **38.00** for amateurs. A similar pattern appears in Montreal (76.64 vs 45.54). In both cities, not only are medians higher for professionals, but the interquartile ranges (IQR) are also shifted upward, indicating a consistent positioning in higher price segments.

From a business perspective, this suggests that professional hosts in these markets operate with a **premium or standardized pricing strategy**, likely supported by better locations, higher-quality listings, or more consistent service.

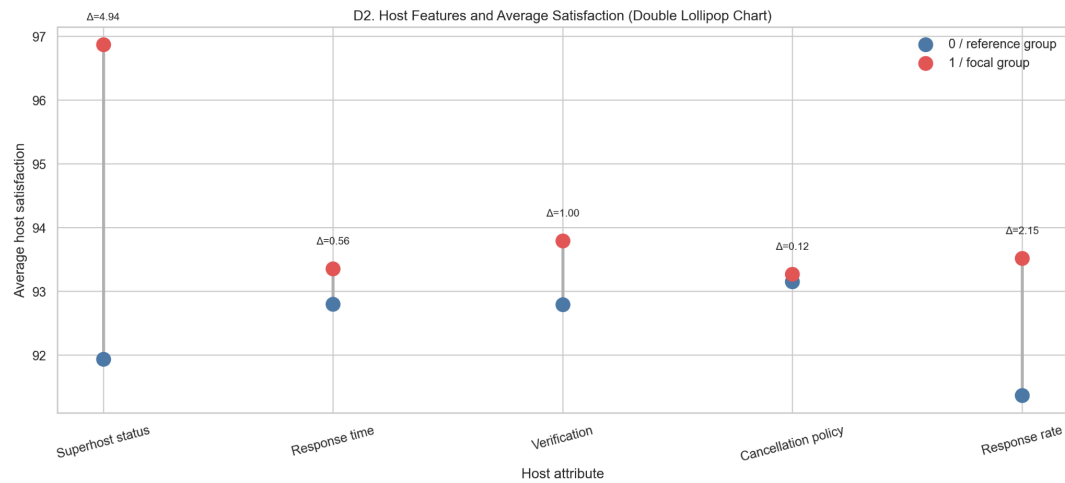
Hong Kong presents a contrasting pattern. Amateur hosts show a higher median (**358.79**) than professionals (**257.00**), along with a much wider spread and extreme maximum values (up to **5000.50**). These high values are not necessarily errors but likely reflect **real high-end listings**, influenced by factors such as location or property uniqueness.

The graph also highlights substantial variability within each group. Wide ranges and visible outliers suggest that **host type alone does not explain pricing**. Instead, it interacts with other variables such as location and property characteristics.

Overall, the results indicate that professional status does not universally lead to higher prices. Its effect depends on local market dynamics, making it important to interpret pricing strategies within each city rather than across them.

D2 Host Features Impact on Satisfaction

How Host characteristics influence customer satisfaction



(Graph: Double lollipop chart of satisfaction differences by host features)

This analysis evaluates how host characteristics influence customer satisfaction, measured using `review_scores_rating`. Satisfaction was aggregated at host level using `host_id`, averaging scores across listings.

Five features were tested: `host_is_superhost`, `host_response_time`, `host_identity_verified`, `host_response_rate`, and `cancellation_policy`, converted into binary indicators where needed (e.g., response rate $\geq 90\%$).

The strongest effect comes from `host_is_superhost`, where satisfaction increases from **91.94 to 96.87** (+4.94). This indicates that the Superhost label reflects meaningful differences in service quality.

`host_response_rate` also shows a clear impact (91.37 vs 93.52, +2.15), suggesting that consistent communication is more important than speed.

Other features have smaller effects: `host_identity_verified` (+1.00) and `host_response_time` (+0.56), while `cancellation_policy` shows almost no influence (+0.12).

The lollipop chart highlights that only a few features create significant differences, reinforcing that satisfaction is mainly driven by **reliability and trust-related metrics**.

How indexing influence performance

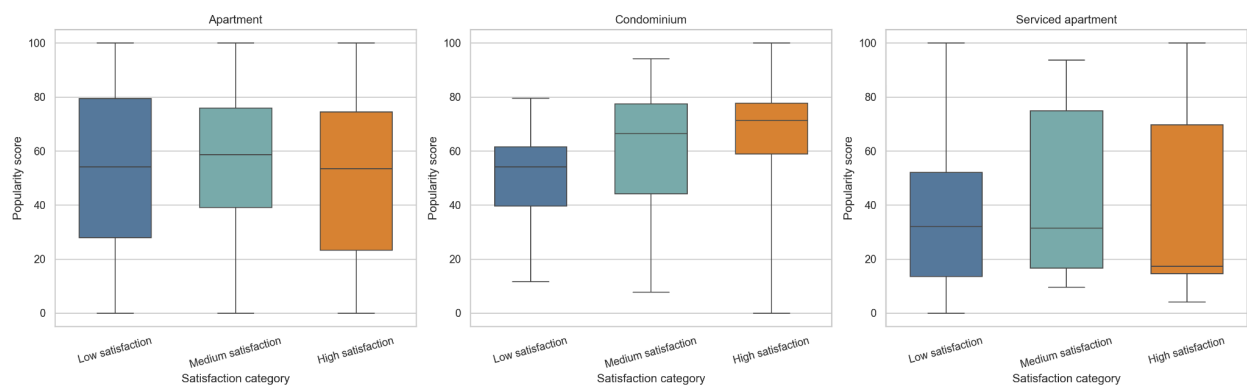
From a performance perspective, optimizing the aggregation pipeline significantly improved efficiency:

Scenario	Stage Pattern	Execution Time (ms)	Docs Examined	Keys Examined	Indexes Used
Before optimization	\$cursor → \$lookup → \$addFields → \$group	15832	28,363,379	0	None
After optimization	\$cursor → \$group → \$sort → \$group → \$addFields	63	9,352	9,352	idx_hosts_host_id_unique, idx_listings_host_id

The improvement comes from restructuring the query and using indexes on `host_id`, reducing unnecessary document scans.

D3 Property Popularity and Satisfaction

D3. Popularity Score by Satisfaction Category for the Three Most Common Professional-Host Property Types



(Graph: Box plots of `popularity_score` by satisfaction category and property type)

This analysis explores the relationship between satisfaction (`review_scores_rating`) and demand, approximated using availability data.

Popularity was measured through a derived `popularity_score`, based on unavailability across four horizons: `availability_30`, `availability_60`, `availability_90`, and `availability_365`.

The score was calculated as:

$$0.4 \times u_{30} + 0.3 \times u_{60} + 0.2 \times u_{90} + 0.1 \times u_{365},$$

where each $u_N = (N - \text{availability}_N) / N$, giving greater weight to short-term demand.

Listings were grouped into satisfaction categories using rating thresholds: **Low (<90)**, **Medium (90–96)**, **High (≥96)**. Median popularity increases from **48.75 (Low)** to **58.08 (Medium)** and remains similar at **57.53 (High)**, suggesting a positive but non-linear relationship.

When segmented by `property_type`, different patterns emerge. Apartments show only a weak relationship between satisfaction and popularity, indicating that other factors (e.g., price or location) may dominate. Condominiums display a clear upward trend (54.19 → 66.44 → 71.44), suggesting that **higher satisfaction directly translates into higher demand** in this segment.

Serviced apartments show a declining trend, but this may be influenced by smaller sample size or different usage patterns.

The box plots reveal overlapping distributions, meaning some low-rated listings still achieve high popularity. This highlights that demand is multi-factorial and not solely driven by satisfaction.

A key limitation is that unavailability may include blocked dates, not only bookings. Despite this, the metric provides a useful approximation of demand.

Overall, the analysis suggests that satisfaction contributes to popularity, but its impact varies depending on property type and market conditions.